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SUPERSEDING
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FEDERAL SPECIFICATION

ANODES, COPPER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for copper anodes used in electroplating baths.

1.2 Classification.

1.2.1 Types. Copper anodes shall be of the following types, as specified (see 6.2):

Type I - Non-deoxidized
Type II - Oxygen free
Type III - Deoxidized, phosphorous bearing

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

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Federal Standards:

Fed. Std. No. 123	-	Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151	-	Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD0129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E-3	-	Standard Methods of Preparation of Metallographic Specimens

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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3. REQUIREMENTS

3.1 Materials. Anodes shall be of solid copper. They shall be either cast, extruded, rolled or forged to shape or cut from electrolytic cathodes and sheets, castings or rolled plate.

3.2 Size, shape and methods of suspension. The size, shape and method of suspension (when applicable) of the anodes shall be as specified (see 6.2).

3.2.1 Ball anodes. Ball anodes shall have a diameter of approximately 2 inches and weigh not less than 1-1/4 pounds.

3.2.2 Sheared bars and shapes. Unless otherwise specified, sheared bars and shapes with none, one or both ends hemispherical are acceptable provided the overall length does not exceed six inches. For Type II copper anodes only, cut bars, shapes or slugs with a maximum length of six inches and a maximum diameter of two inches will be acceptable.

3.2.3 Bar anodes. Bar anodes shall be furnished in lengths of 18 to 108 inches as specified. Unless otherwise specified, the approximate cross sectional dimensions and the minimum weight per linear inch for various forms of bar anodes and cut shapes shall be as shown in Table I. Unless otherwise specified, flat cut shapes shall be approximately 1/2 inch in thickness.

TABLE I

DIMENSION AND WEIGHTS

Form	Dimension (approximate)	Minimum weight per inch of length-pounds
Round Bar Anodes	3 inches diameter	2.25
	2 inches diameter	1.00
Oval Bar Anodes	3 inches by 2 inches	1.50
	3-1/8 inches by 1-1/2 inches	1.25
Flat Bar Anodes	6 inches by 1-1/2 inches	2.90
	3 inches by 1 inch	1.00
Cut Shapes	2 inches diameter by 3 inches	1.00
	2 inches diameter by 4 inches	1.00
	2 inches diameter by 6 inches	1.00
	6 inches by 3 inches	3.80
	4 inches by 1-1/2 inches	2.00
	1-1/2 inches by 1-1/2 inches	0.70
	1 inch by 1 inch	0.32

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3.3 Chemical composition. Unless otherwise specified in the contract or order (see 6.2), the copper anodes shall meet the chemical requirements shown in Table II (see 6.4).

TABLE II

CHEMICAL COMPOSITION - PERCENT 3/

Element	Type I Non-deoxidized	Type II Oxygen-free	Type III Deoxidized, phosphorous Bearing
Copper 1/	99.90 Min	99.95 Min	99.90 Min
Iron	0.003 Max	0.002 Max	0.003 Max
Sulfur	0.003 Max	0.003 Max	0.003 Max
Lead	0.003 Max	0.002 Max	0.002 Max
Antimony	0.002 Max	0.002 Max	0.002 Max
Nickel	0.003 Max	0.002 Max	0.002 Max
Arsenic	0.001 Max	0.001 Max	0.001 Max
Total, other Metallic elements 2/	0.01 Max	0.01 Max	0.01 Max
Oxygen	0.045 Max	--	--
Phosphorous	--	0.001 Max	0.015 to 0.040

1/ Silver is counted as copper.
2/ Total is the sum of other elements (bismuth, cadmium, manganese, mercury, tin, selenium, tellurium, and zinc) present as impurity in the sample.
3/ Analysis shall be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that these other elements are not present in excess of the limits specified.

3.4 Cuprous oxide. Type II and III copper anodes shall be free from cuprous oxide when tested in accordance with the microscopic test as specified in 4.5.2.

3.5 Identification. Each anode, except ball anodes, shall be permanently identified with the following information:

(a) "Anode, Copper"
(b) Specification Number
(c) Type Classification

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3.5.1 Ball anodes. Each ball anode shall be permanently marked with the word "Copper" or the symbol "Cu", either as the result of casting or otherwise imprinting the selected identifying mark.

3.6 Workmanship. Anodes shall be clean and substantially free from cracks, warps, casting laps, inclusions, porosity, ragged edges, surface films such as rolling skin, and other defects which may adversely affect uniform corrosion in service. Anodes shall not be broken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order (see 6.2), a lot shall consist of all anodes of one type and shape in a single shipment. When practicable, each lot shall be made up of anodes of the same melt, size and weight.

4.3 Quality conformance sampling.

4.3.1 For visual examination and non destructive tests. Unless otherwise specified (see 6.2), samples for visual examination and non destructive tests shall be selected at random from each lot in accordance with MIL-STD-105, Level II, Acceptable Quality Level (AQL) 1.5 percent defective.

4.3.2 For chemical analysis and microscopic examination. Sample anodes shall be selected as follows:

(a) For ball anodes or sheared bars and shapes not exceeding six inches in length, at least 4 sample anodes shall be selected from each lot or from each melt if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 8 anodes shall be selected at random from each lot.

(b) For shapes other than balls or sheared bars and shapes not exceeding six inches in length, at least 2 sample anodes shall be selected at random from

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each lot or from each melt if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 4 anodes shall be selected at random from each lot.

4.3.2.1 Preparation for chemical analysis. A composite sample of at least 6 ounces of chips or drillings shall be obtained by drilling, sawing, milling or machining the sample anodes of 4.3.2 in accordance with Fed. Test Method Std. No. 151, Method 111. No lubricant shall be used in the operation. The sample shall be free from dirt, oil, grit and other foreign matter. Chips or drillings shall be treated with a magnet to remove any steel introduced in the preparation of the sample. Anodes which have been drilled or machined for chemical analysis may be included in the lot for shipment.

4.3.2.2 Preparation for spectrochemical analysis. A composite sample shall be prepared as detailed in 4.3.2.1 or rod electrodes shall be prepared from each of the sample anodes of 4.3.2 in accordance with Fed. Test Method Std. No. 151, Method 112. The rod samples shall be prepared by conventional methods, suitable for the correct applications and usages of the techniques employed. No lubricant shall be used in the preparation operation. The sample shall be free from dirt, oil, grit and other foreign matter. Anodes, except for the ball and small slug types, from which electrodes have been obtained may be included in the lot for shipment.

4.3.2.3 For microscopic examination. The Types II and III anodes that have been selected in accordance with 4.3.2, and prepared in accordance with 4.3.2.1 or 4.3.2.2, shall be used for the determination of the presence of cuprous oxide.

4.4 Quality conformance examination.

4.4.1 Anodes. Sample anodes, selected in accordance with 4.3.1 shall be visually inspected, measured and weighed for conformance to the requirements of 3.2, 3.5 and 3.6. If the number of defective items in any sampling exceeds the acceptance number for the specified sample size, the lot represented by the sample shall be rejected.

4.4.2 Packing and marking. The packing and marking of anodes furnished under this specification shall be inspected for conformance to the applicable requirements of Section 5.

4.5 Quality conformance testing.

4.5.1 Chemical composition. Sample anodes, selected in accordance with 4.3.2, shall be tested in accordance with Method 111 or Method 112 of Fed. Test Method Std. No. 151 to determine conformance to the chemical

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requirements of Table II. If the sample fails to conform to the requirements of Table II, the lot shall be rejected. Standard methods of analysis shall be used (see 6.).

4.5.2 Cuprous oxide. Sample anodes selected in accordance with 4.3.2.3 shall be tested for the presence of cuprous oxide for compliance with 3.4. Specimens shall be prepared in accordance with ASTM E3. The unetched specimen shall be microscopically examined at 75 diameters magnification in order to detect the presence of cuprous oxide: Any sample found to contain cuprous oxide shall constitute failure and rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Packing. When practicable, anodes of the same size and shape, shall be packed together. Unless otherwise specified, the gross weight of each box shall not exceed approximately 200 pounds (see 6.2).

5.1.1 Level A. Anodes shall be packed in overseas type boxes in accordance with PPP-B-601 or PPP-B-621.

5.1.2 Level B. Anodes shall be packed in domestic type boxes in accordance with PPP-B-585, PPP-B-601 or PPP-B-621.

5.1.3 Level C. Anodes shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Containers shall meet, as a minimum, the requirements of rules and regulations applicable to the mode of transportation selected.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

5.2.3 Special marking. Each container shall be marked with the following:

(a) Specification Number
(b) Nomenclature of anode, as applicable
(c) Name of contractor
(d) Contract Number or Order Number
(e) Quantity
(f) Gross Weight

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6. NOTES

6.1 Intended use. Copper anodes are intended for use in copper-plating baths.

6.1.1 Performance. Cast, forged, rolled or electrolytic copper anodes are satisfactory in cyanide baths. Cast ball or cast oval bars are usually used with a Rochelle cyanide bath whereas rolled or cast bars are used in high-concentration sodium cyanide baths. In the sodium cyanide bath, the anode is often bagged to form an anode compartment. Cast and rolled bar anodes are satisfactory in acid copper sulfate baths but there is some evidence that electrolytic copper anodes are less satisfactory in this type bath. Withe the acid copper sulfate bath, cast anodes are unsuitable for use in electroforming because they contain considerable copper oxide. If about one-eight of the copper anode surface area is replaced with lead anodes, the metallic copper and the sulfuric acid in the bath will be held almost constant. Rolled or sheet electrolytic copper anodes are satisfactory in pyrophosphate baths. Anodes with this type of bath should not be bagged and should be free of tin, nickel, silver and lead because these elements promote roughness of deposits. In fluoborate baths, rolled or cast bars, or electrolytic sheet anodes are satisfactory. When thick smooth deposits are required to be produced from fluoborate baths, the anodes should be contained in vinyl-yarn bags to prevent the reddish-brown sludge that is common to acid copper electrolytes from entering the solution. There is also some evidence that oxygen free anodes produce less sludge in the cyanide bath and oxygen free, phosphorous bearing anodes are more satisfactory in this respect in acid copper baths. Phosphorized anodes in copper sulphate baths create a heavy, tenacious film that retains the fine particles of sludge which would otherwise cause particle roughness.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.

(b) Type and nomenclature as applicable (see 1.2.1).

(c) Size, shape and method of suspension (when applicable) (see 3.2).

(d) Chemical composition, if different from Table II (see 3.3).

(e) Size of lot, if different from 4.2.

(f) Sampling procedure, if different from 4.3.1.

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(g) Gross weight of packing, if different from 5.1.

(h) Applicable level of packing (see 5.1.1, 5.1.2 and 5.1.3).

(i) Additional marking, if necessary (see 5.2).

6.3 Methods of chemical analysis. It is intended that the chemical analysis be conducted so that no particular method of analysis be specified. As a guide, copper contents in anodes may be determined in accordance with ASTM E-53, Standard Method for Chemical Analysis of Copper (Electrolytic Determination of Copper). Metallic impurities in copper anodes may be determined by photometric methods in accordance with ASTM E62, Standard Photometric Methods for Chemical Analysis of Copper and Copper-Base Alloys or by spectrographic methods in accordance with ASTM E414, Standard Method for Spectro Chemical Analysis of Copper by the Briquet D-C Arc Technique. For determination of oxygen in the anodes, the conductometric method or the vacuum fusion method should be used.

6.4 Commercial designations. Table III lists the standard numerical designation system and the commonly accepted trade names for the types of anodes specified herein. The compositional limits of the referenced Copper Development Association Number may not be identical to those listed in Table II. Though not directly interchangeable in these cases, such alloys may be considered approximately equivalent.

TABLE III

COMMERCIAL DESIGNATION

Type	Copper Alloy Number	Commonly Accepted Trade Name
I - Non-oxidized	110	Electrolytic Tough Pitch
II - Oxygen free	101 and 102	Oxygen Free Certified and Oxygen Free
III - Deoxidized, phosphorous bearing	122	Phosphorous Deoxidized High Residual Phosphorous

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Military Custodians
Army - WC
Navy - AS
Air Force - 84

Review Activities
Army - None
Navy - SH
Air Force - None
Defense Supply Agency - GS

User Activities
Army - GL, WC
Navy - OS
Air Force - None

Civil Agencies Interest
GSA - FSS

Preparing Activity
Navy - AS
(Proj. No. 3426-0008)

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